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Shifa Tameer-e-Millat University

Project Proposal

for

Lostify AI

Version 1.0

By

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- A- Artificial Intelligence & Machine Learning
- B- Web Application
- C- Mobile Application

Abstract

Lostify AI is a proposed mobile and web-based recovery platform designed to help citizens report and recover missing persons, lost pets and lost items through one secure digital system. Reporting is often fragmented across social-media posts, manual police processes, community groups and separate pet or device services, which delays the circulation and verification of useful information. Lostify AI addresses this gap through structured reports, location-aware alerts, role-based access for citizens, volunteers, NGOs, police and administrators, and AI-assisted image similarity, object detection and OCR support. AI suggestions are not treated as final identity decisions; high-risk matches require human review. The project aims to improve the speed, reach, traceability and safety of recovery coordination while protecting personal information.

1. Introduction

Computer vision and artificial intelligence have significantly transformed the way digital systems analyze images, videos, and visual information to support intelligent decision-making. These technologies are increasingly applied in public safety, healthcare, transportation, surveillance, and smart city applications to automate complex tasks and improve operational efficiency. The widespread use of smartphones, cloud computing, and geospatial services has enabled the development of real-time reporting and location-aware applications for emergency and public assistance. Modern mobile and web technologies such as Flutter, Node.js, PostgreSQL, Google Maps Platform, and Firebase further support the development of scalable and cross-platform solutions for large communities [4]–[8]. Additionally, advances in computer vision models, including **object detection, face recognition, and optical character recognition (OCR)**, have made it possible to identify people, animals, and objects with greater accuracy and speed [9], [11], [12], [17]. These technological advancements provide a strong foundation for developing intelligent recovery systems capable of assisting users during missing-person and lost-property incidents.

Several digital platforms currently provide recovery services for different types of missing entities. The **National Center for Missing & Exploited Children (NCMEC)** supports **missing-child** reporting and coordinates awareness campaigns with law enforcement agencies to improve child recovery efforts [1]. Google Find Hub enables users to locate, lock, secure, or erase registered Android devices linked to their Google accounts, helping users recover or protect lost mobile devices [2]. Petco Love Lost provides AI-assisted photo matching and community reporting to improve the recovery of lost pets through image comparison technology [3]. These systems have demonstrated that digital reporting, image analysis, and location-based services can significantly improve recovery processes within their respective domains. However, each solution focuses on a specific category and does not provide a common platform capable of supporting multiple recovery scenarios.

Despite their effectiveness, existing systems present several limitations and research gaps. Most platforms are restricted to a single recovery category, such as missing children, pets, or registered electronic devices, making users depend on multiple independent applications [1]–[3]. Social media platforms are also widely used for reporting missing persons and lost items, but they lack structured data collection, verification mechanisms, and reliable search capabilities, often resulting in duplicated or misleading information. Existing solutions rarely integrate AI-powered image matching, geospatial alerts, community participation, and secure verification within one unified workflow. Furthermore, limited interoperability, inconsistent moderation, and insufficient audit mechanisms reduce the efficiency and trustworthiness of current recovery systems. These challenges highlight the need for a secure, AI-assisted, and comprehensive platform capable of supporting diverse recovery cases through a standardized process.

To address these limitations, this project proposes **Lostify AI**, an intelligent recovery platform that unifies reporting and recovery processes for missing persons, lost pets, and personal belongings within a single system. The platform enables users to submit structured reports, upload images, specify last-known locations, and receive nearby notifications based on geographic information. Artificial intelligence is incorporated to generate similarity-based suggestions using computer vision techniques, while human verification remains mandatory before any recovery decision is confirmed. Community members can contribute verified sightings and evidence, improving collaboration and increasing the likelihood of successful recovery. By integrating AI-assisted matching, location-aware services, secure verification, and community participation, Lostify AI provides a comprehensive and scalable solution that complements existing recovery services rather than replacing official authorities [4], [9]–[12].

The proposed system consists of several integrated technical modules that collectively support the recovery workflow. The citizen mobile application, developed using Flutter, enables **users to register, submit reports, upload evidence, and monitor case progress** [6]. **The backend services, implemented with Node.js, Express, and PostgreSQL, securely manage authentication, case records, role-based access, and spatial information** [7], [8]. AI-assisted services utilize **computer vision algorithms, including YOLO for object detection, ArcFace and FaceNet for facial recognition, and PP-OCR for extracting textual information from images** [9], [11], [12], [17]. Google Maps Platform provides location visualization and geospatial services, while Firebase supports real-time notifications and communication between users [4], [5]. Security requirements are further strengthened using the OWASP Application Security Verification Standard to ensure data protection, secure access, and responsible AI deployment [10].

In conclusion, Lostify AI aims to improve the efficiency, accessibility, and reliability of recovery operations by integrating artificial intelligence, computer vision, geospatial technologies, and community collaboration into a unified platform. Unlike existing systems that address only specific recovery scenarios, the proposed solution supports multiple categories within a single standardized workflow while maintaining human oversight for sensitive decisions. The project combines modern software engineering practices with advanced AI technologies to enhance recovery speed, reduce duplicate reporting, and improve public participation. Its modular architecture also supports future expansion through additional AI models, partner organizations, and advanced analytics. Consequently, Lostify AI represents a practical and innovative contribution toward the development

of intelligent recovery systems for missing persons, lost pets, and personal belongings while promoting secure, transparent, and accountable information management.

2. Problem Statement

Missing-person, lost-pet and lost-item cases are currently handled through separate and often informal channels. Families and owners repeatedly post the same information to social media, contact authorities, visit shelters or organizations, and respond to unverified messages. This creates duplicated reports, incomplete evidence, delayed alerts and a risk of false claims or scams. Current tools usually focus on one category only and do not provide one common workflow for reporting, matching, alerting, verification and recovery closure. A secure, role-based and AI-assisted platform is required to centralize reliable case information, suggest possible matches, notify relevant nearby users and maintain a traceable handover process.

3. Proposed Problem Solution

Lostify AI will provide a unified application for the reporting and recovery of missing persons, lost pets and lost items. A user will select a case category, enter verified contact information, upload permitted images, describe the case and pin the last-seen location. The platform will store the case securely, distribute controlled nearby notifications, accept community sightings and use AI services to rank visually or textually similar records. AI outputs will be presented as suggestions to reduce search effort; final verification and case closure remain the responsibility of authorized users and relevant authorities.

3.1 Objectives

- **BO-1:** Provide a structured, role-based reporting process for missing persons, pets and items.
- **BO-2:** Reduce the time needed to circulate verified cases through location-aware alerts and shareable case pages.
- **BO-3:** Support investigation with AI-assisted image similarity, object detection and OCR without replacing human verification.
- **BO-4:** Improve claim safety through evidence storage, owner challenge questions, audit logs and controlled case access.
- **BO-5:** Provide authorized dashboards for case monitoring, recovery-rate reporting and AI performance review.

4. Related System Analysis/Literature Review

Existing solutions provide valuable recovery features but are primarily designed for specific domains rather than a unified recovery platform. **The National Center for Missing & Exploited Children (NCMEC)** focuses on missing-child reporting, public awareness, and coordination with law enforcement agencies to support child recovery efforts [1]. **Google Find Hub** enables users to locate, secure, lock, or erase registered Android devices, but its functionality is limited to enrolled devices linked to a Google account [2]. **Petco Love Lost** offers AI-powered photo matching and community reporting for lost and found pets, providing an effective solution specifically for pet recovery [3]. Although these platforms address important recovery scenarios, each operates within a separate domain. **Lostify AI** is proposed as a unified recovery platform that complements rather than replaces official authorities or specialized services by integrating structured case management for lost people, pets, and personal belongings. The proposed system combines AI-assisted matching using computer vision technologies such as YOLO and face recognition techniques [9], [11], [12], location-aware reporting through Google Maps services [4], community sighting reports, local alert notifications, and accountable human verification to improve recovery efficiency across multiple categories while maintaining transparency and security [10].

Table 1 Related System Analysis with proposed project solution

Application Name	Weakness	Proposed Project Solution
NCMEC / AMBER-style services	Focused on missing children and official processes.	Adds multi-category reporting and community-sighting workflows.
Google Find Hub	Locates enrolled Android devices only.	Adds public lost-item reports, QR registration and claim verification.
Petco Love Lost	Provides pet-focused reporting and photo matching.	Extends structured matching across pets, people and items.
Social media groups	Unstructured posts and weak verification.	Creates standard fields, controlled access and audit logs.
Paw Boost	Limited to lost pet recovery.	Provides one platform for pets, people, and lost valuables.
Apple Find My	Works only within the Apple ecosystem.	Supports people, pets, and items across all platforms.

5. Vision Statement

For families, owners, volunteers and partner organizations who need a faster and more trustworthy way to coordinate recovery, Lostify AI is a secure AI-assisted recovery platform that records cases, surfaces relevant leads and enables controlled community action. Unlike disconnected social posts and category-specific tools, our product unifies missing-person, lost-pet and lost-item workflows with location-aware alerts, evidence handling, ownership verification and role-based oversight.

6. Scope

The scope includes a citizen-facing mobile application and a web-based administration and partner portal. The mobile application supports account creation, report submission, photo and document upload, map-based last-seen locations, nearby case browsing, sighting submission, QR scanning, controlled messaging and case-status tracking. The portal allows verified NGO, police and administrator users to review reports, manage verification queues, issue approved broadcasts, examine AI suggestions and close verified recoveries. The initial FYP prototype uses test data and does not claim official police integration, live emergency dispatch, automatic legal identification or mandatory biometric verification. Automated results are suggestions that require human review.

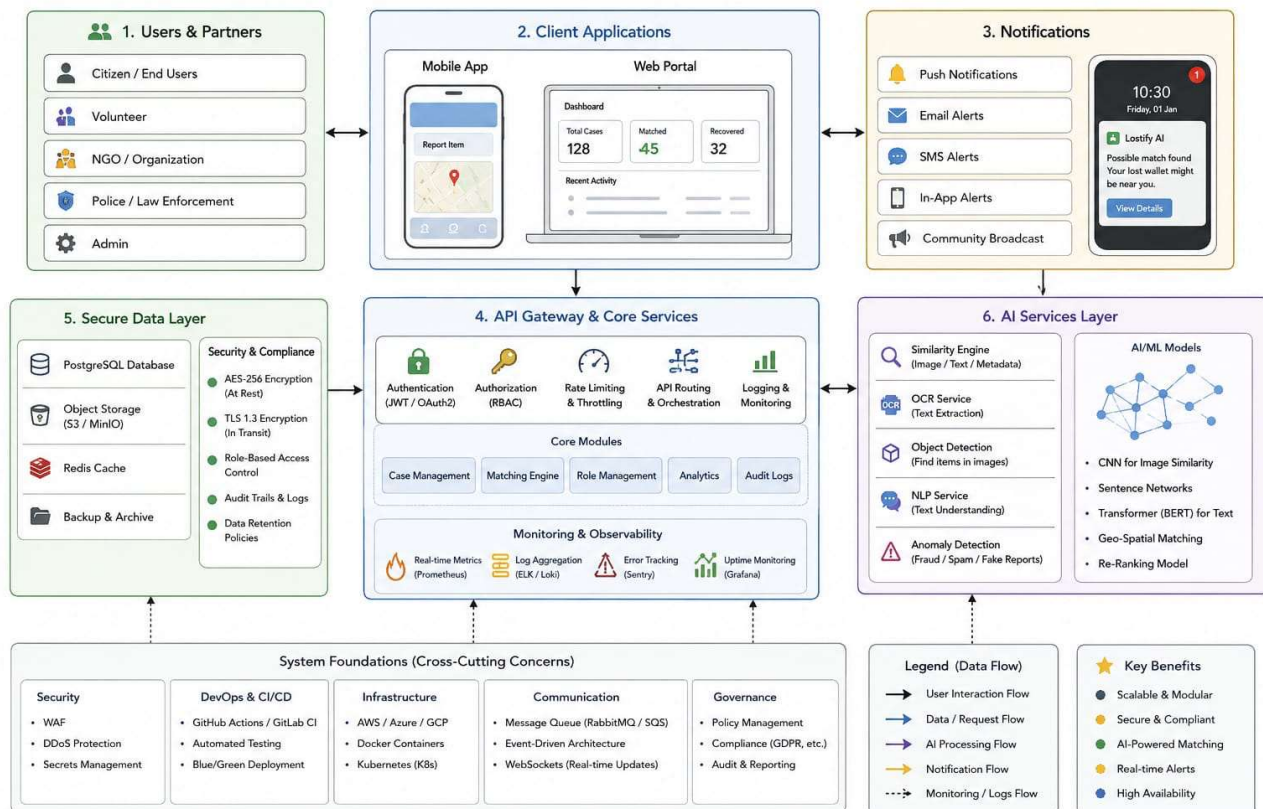


Figure 1: Proposed high-level architecture of Lostify AI.

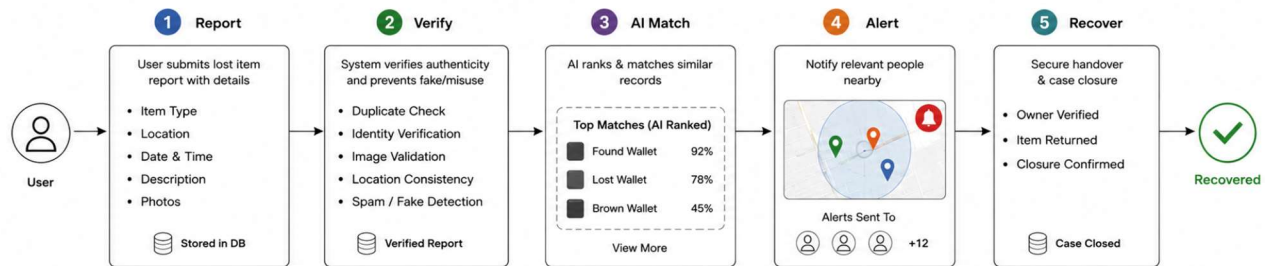


Figure 2: Controlled case recovery workflow.

7. Modules

7.1 Citizen Mobile Application

- **FE-1:** Register and manage a verified user profile.
- **FE-2:** Report a missing person, lost pet or lost item with images, description and last-seen location.
- **FE-3:** View nearby approved cases, alerts and case timelines.
- **FE-4:** Submit a sighting or anonymous tip with time, location and permitted media.
- **FE-5:** Scan QR tags for registered pets or valuables.

7.2 AI-Assisted Matching Services

- **FE-6:** Create image embeddings to return ranked visually similar records.
- **FE-7:** Detect selected objects, clothing or collar cues in permitted images.
- **FE-8:** Extract serial numbers, IMEI or document text through OCR.
- **FE-9:** Flag likely duplicate reports and prioritize high-risk cases.
- **FE-10:** Show confidence as a non-final suggestion with human-review requirement.

7.3 Partner and Administration Portal

- **FE-11:** Review reports, verification status, evidence and audit history.
- **FE-12:** Manage roles for administrators, NGOs, volunteers and authorized police users.
- **FE-13:** Approve broadcasts and view case heatmaps and recovery outcomes.
- **FE-14:** Export authorized case-summary PDFs.
- **FE-15:** Monitor AI response time and quality feedback.

7.4 Security and Communication Module

- **FE-16:** Role-based access control and protected sessions.
- **FE-17:** Secure storage for evidence, images and documents.

- **FE-18:** Nearby push, email or SMS alerts according to consent.
- **FE-19:** Owner claim questions and verified handover confirmation.
- **FE-20:** Audit logs for sensitive access and edits.

8. System Limitations/Constraints

- **LI-1:** The FYP prototype will not connect directly to live police, hospital, shelter or telecom databases without formal approval.
- **LI-2:** AI similarity and OCR outputs can be inaccurate and must be reviewed by a human.
- **LI-3:** The system will not provide real-time tracking unless a user voluntarily shares location with explicit consent.
- **LI-4:** Emergency cases must still be reported through relevant official emergency channels.
- **LI-5:** Sensitive personal and biometric data requires consent, restricted access, retention limits and secure storage.
- **LI-6:** Location notifications depend on permissions, connectivity and device accuracy

9. Data Gathering Approach

Requirements will be gathered through semi-structured interviews with potential users such as students, families, pet owners and volunteers; short questionnaires about reporting pain points and privacy expectations; and observation of publicly documented recovery workflows. A prototype usability session will ask participants to complete report, sighting and match-review tasks using dummy data. Secondary research will study official missing-child resources, lost-device guidance, lost-pet services and responsible AI evaluation material. No real case data or sensitive biometric information will be collected for the academic prototype without explicit approval and consent.

10. Tools and Technologies

Table 2: Tools and Technologies for Proposed Project

Layer	Tool / Technology	Version	Rationale
Mobile client	Flutter / Dart	Flutter 3.x / Dart 3.x	Cross-platform citizen application.
Web portal	React + TypeScript	React 18+	Partner and administrator dashboard.
Backend	Node.js + Express	Node 22 LTS	REST APIs and authentication.

Database	PostgreSQL + PostGIS	PostgreSQL 16+	Case data and spatial queries.
AI service	Python + FastAPI + PyTorch	Python 3.11+	Model inference APIs.
Computer vision	OpenCV/ embeddings / OCR / Keras	Prototype models	Object cues, similarity and extraction.
Maps and alerts	Google Maps SDK / Firebase	Current SDKs	Maps and notifications.

11. Project Stakeholders and Roles

Table 3 Project Stakeholders for Proposed Project

Stakeholder	Role and Responsibility
Citizens / families / owners	Create reports, supply accurate information, receive updates and confirm recovery.
Volunteers / community members	View approved nearby cases and submit sightings or tips.
NGO and shelter partners	Review authorized cases and contribute intake information when formally partnered.
System administrators	Manage roles, moderation, broadcasts, audit records and analytics.
Supervisor	Reviews scope, ethics, progress and deliverables.

12. Module based Work Division

Table 4 Team Member Work Division for Proposed Project

Student Name	Student Registration Number	Responsibility / Module / Feature

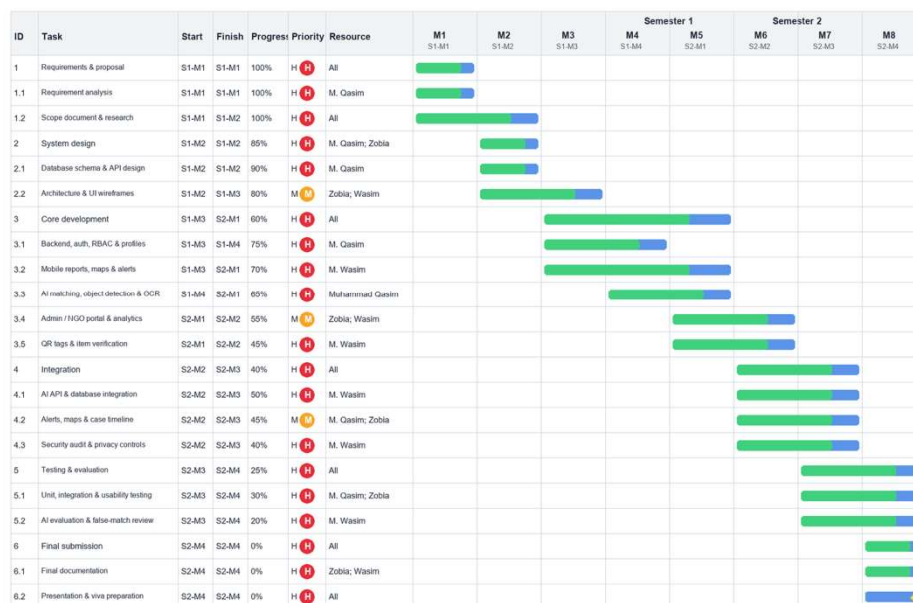
M. Wasim Adil	BSCS-23F-0129	AI services, image similarity, object/OCR experiments, evaluation and analytics support.
Zobia Shahid	BSCS-23F-0143	Frontend mobile app, report forms, maps, notifications UI and mockups.
Muhammad Qasim	BSCS-23F-0126	Backend APIs, authentication, database schema, roles, evidence and portal integration.

13. WBS and Gantt Chart

The planned work is organized to level 2. Design starts after requirements analysis; core development depends on approved designs; AI integration depends on the API and data schema; and final testing depends on integrated modules. Dates are estimated and can be adjusted by the supervisor according to the academic calendar.

WBS with Gantt Chart - 2 Semesters / 8 Months

Team Members: M. Wasim Adil, Zobia Shahid, Muhammad Qasim



Timeline assumes one semester is four months; total planning horizon is 2 semesters / 8 months.

Scope Document for Lostify AI

WBS	Task	Duration	Planned Window	Primary Resource / Dependency
1.0	Requirements and scope	2 weeks	Weeks 1?2	All members
1.1	Interviews, questionnaire and literature review	1 week	Week 1	All members
1.2	Requirements validation and proposal approval	1 week	Week 2	Supervisor dependency
2.0	Architecture, database and UI design	3 weeks	Weeks 3?5	Depends on 1.0
2.1	Wireframes, prototype and data model	2 weeks	Weeks 3?4	Wasim / Zobia
2.2	AI experiment plan and evaluation criteria	1 week	Week 5	Qasim; data model
3.0	Core system development	5 weeks	Weeks 6?10	Depends on 2.0
3.1	Mobile reports, maps and timeline	3 weeks	Weeks 6?8	Wasim
3.2	APIs, database, roles and admin functions	4 weeks	Weeks 6?9	Zobia
3.3	AI matching / OCR prototype service	4 weeks	Weeks 7?10	Qasim; API contract
4.0	Integration and testing	3 weeks	Weeks 11?13	All; depends on 3.0
4.1	Usability, security and AI review	2 weeks	Weeks 11?12	All members
4.2	Final fixes, documentation and demo	1 week	Week 13	All members

14. Mockups

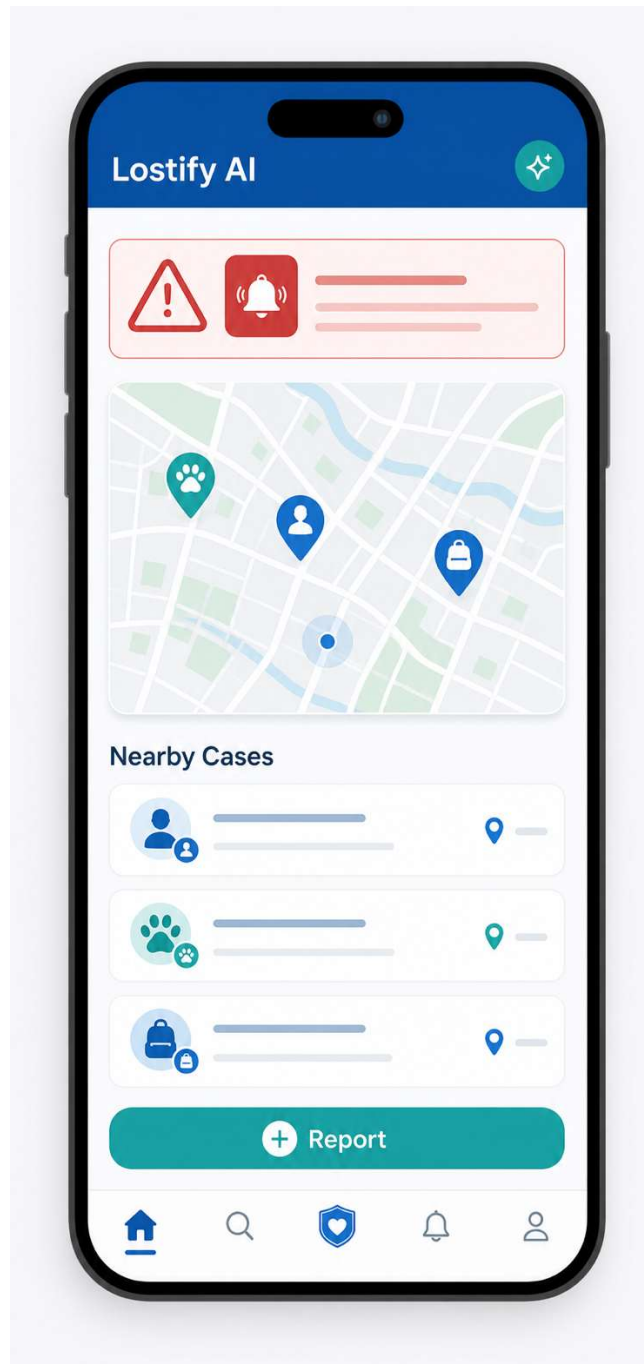


Figure 3: Home and nearby-cases screen with emergency alert, map makers and direct report action

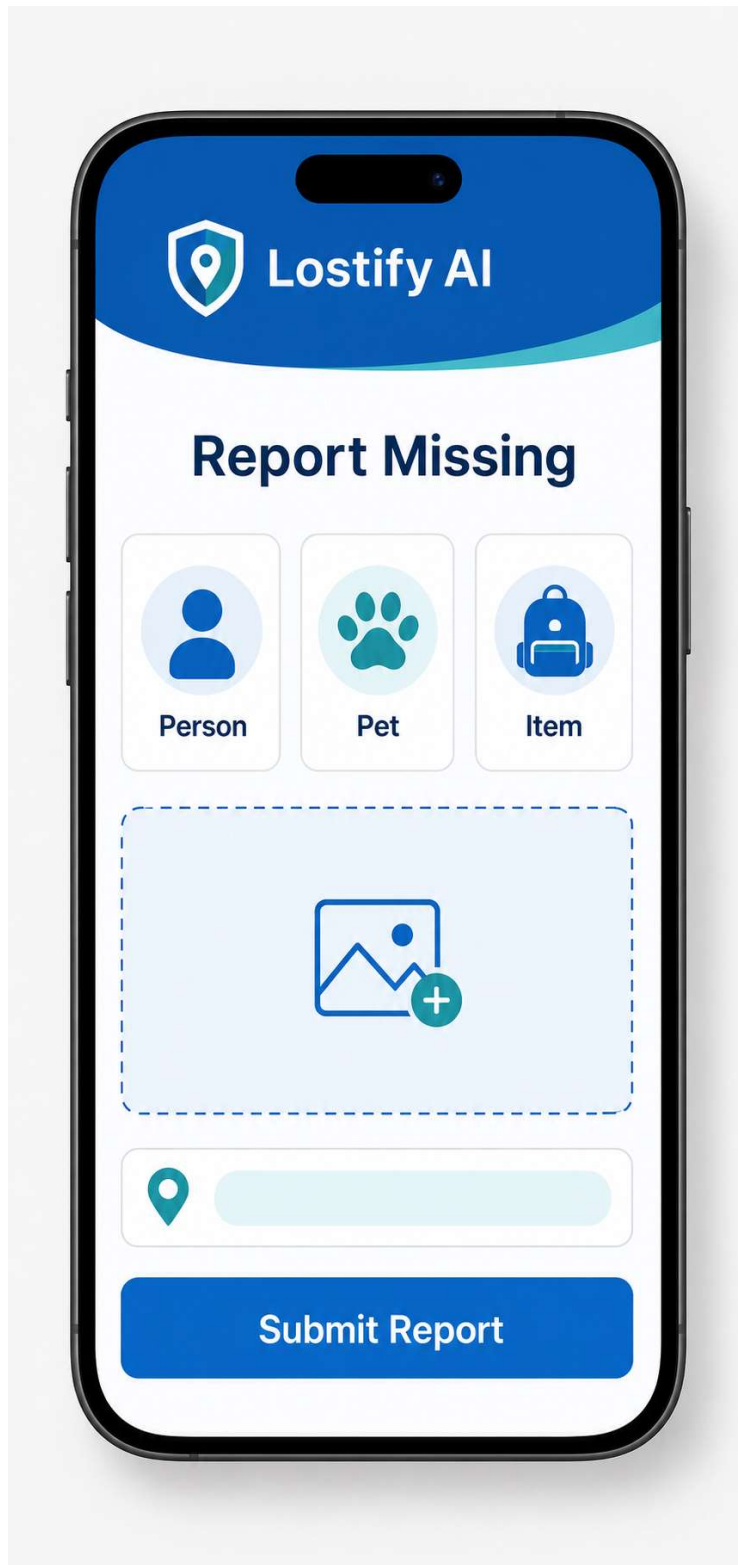


Figure 4: Report-missing screen with category selection, photo upload, location and report submission.

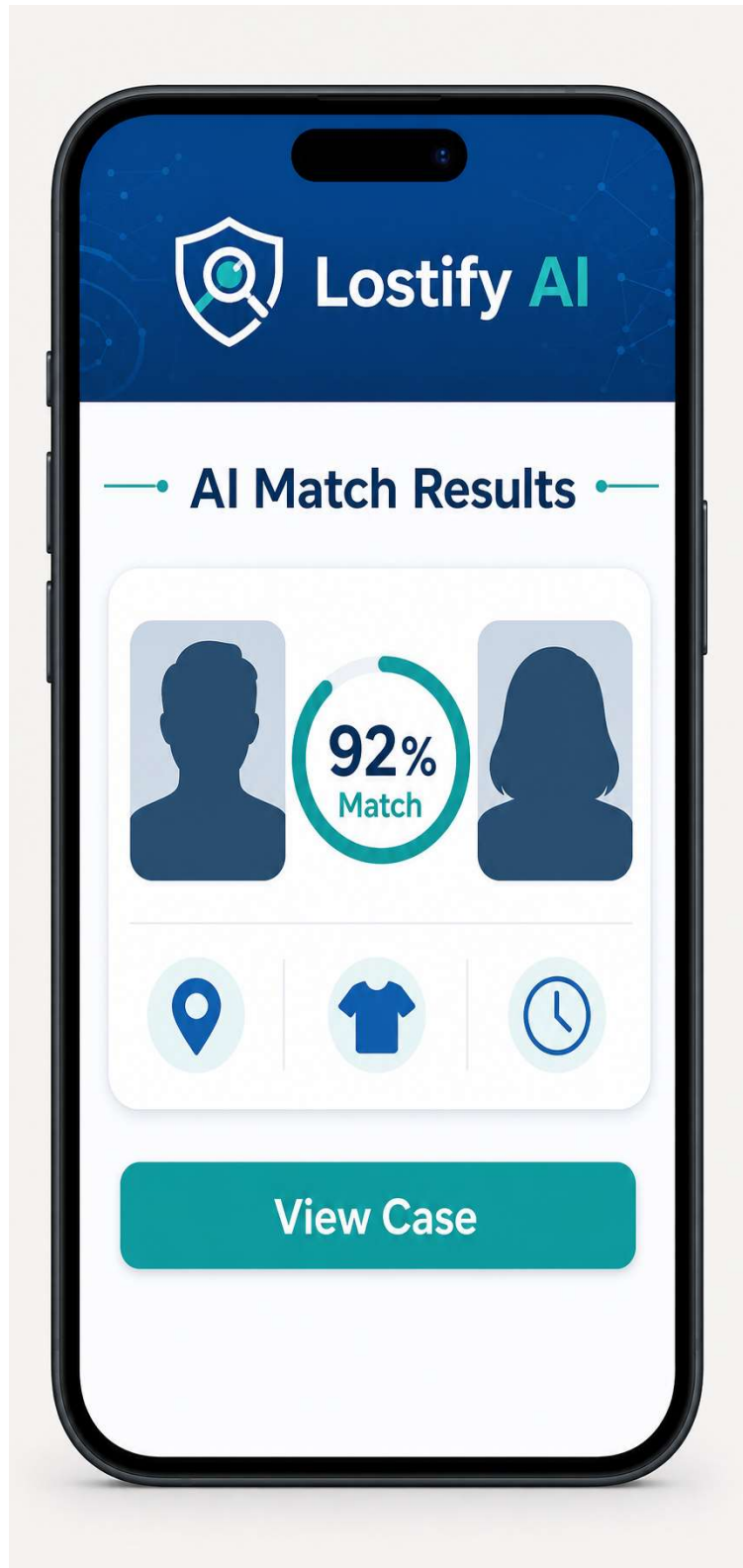


Figure 5: AI match-results screen. The score is an aid for review, not a final identity decision.

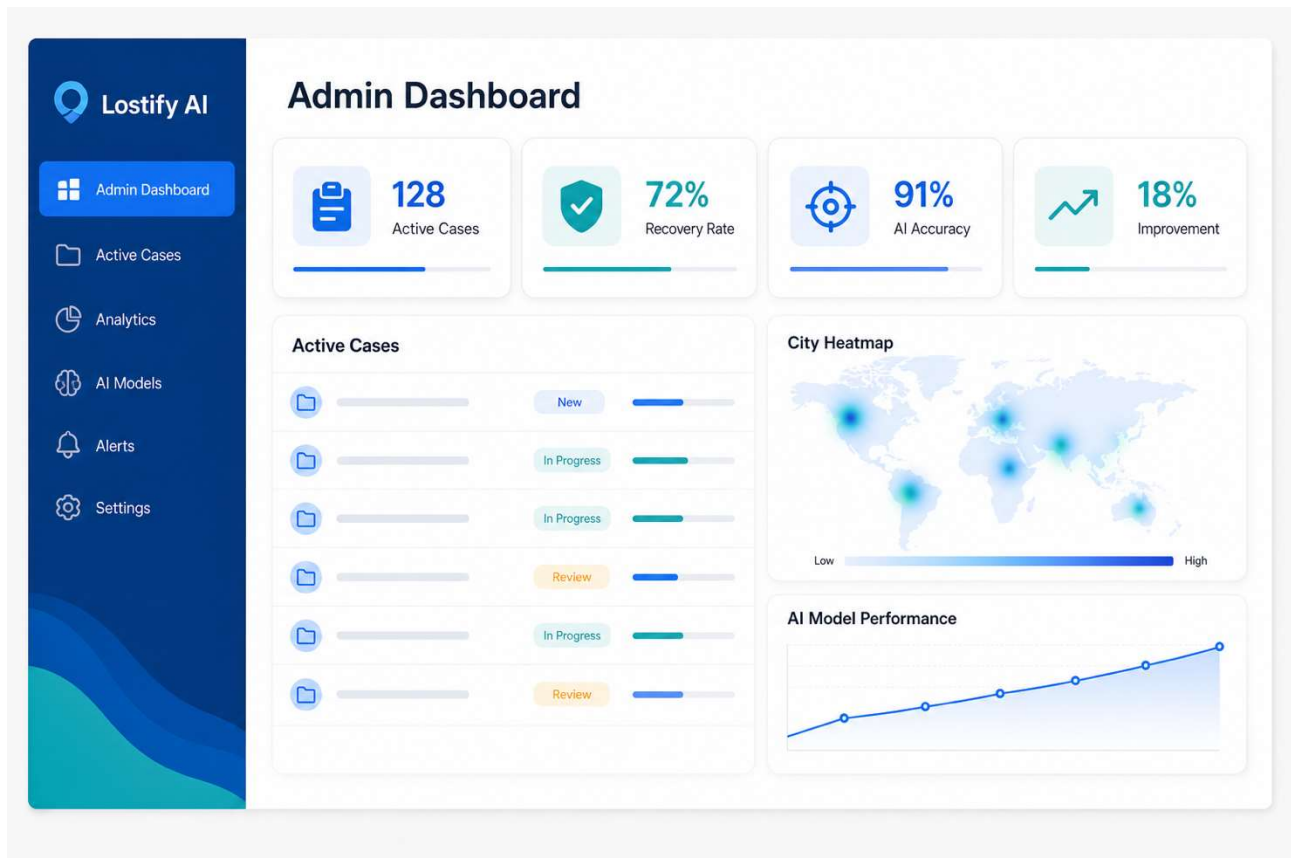


Figure 6: Administration dashboard for active cases, recovery metrics, heatmap and AI monitoring

15. References

If your instructor only wants **World Wide Web** references in the simple format (without "Published", publisher details, etc.), then you can write them like this:

15.1.1 World Wide Web

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